

Database Schema Overview

Core User Management

- **user**
 - id (PK)
 - email, password, role, first_name, last_name
- **student_profile**
 - id (PK), student_id, department, user_id (FK → user.id)
- **instructor_profile**
 - id (PK), user_id (FK → user.id), employee_id, specialization
- **administrator_profile**
 - id (PK), user_id (FK → user.id), employee_id, access_level

Course Structure

- **category**
 - id (PK), name, slug
- **course**
 - id (PK), instructor_id (FK → instructor_profile.id), category_id (FK → category.id), title, description, price, status
- **module**
 - id (PK), course_id (FK → course.id), title
- **lesson**
 - id (PK), module_id (FK → module.id), title, video, pdf
- **announcement**
 - id (PK), title, author_id (FK → user.id), content, visibility, course_id (FK → course.id)

Assignments

- **assignment**
 - id (PK), course_id (FK → course.id), module_id (FK → module.id), title
- **assignment_submission**
 - id (PK), student_id (FK → student_profile.id), graded_by (FK → instructor_profile.id), assignment_id (FK → assignment.id), score

Quizzes

- **quiz**
 - id (PK), title, passing_score, max_attempts, course_id (FK → course.id), module_id (FK → module.id)
- **question**
 - id (PK), quiz_id (FK → quiz.id), question_text, type, points
- **choice**
 - id (PK), choice_text, is_correct, question_id (FK → question.id)
- **quiz_attempt**
 - id (PK), student_id (FK → student_profile.id), score, passed, quiz_id (FK → quiz.id)
- **quiz_answer**
 - id (PK), attempt_id (FK → quiz_attempt.id), is_correct, points_awarded, choice_id (FK → choice.id), question_id (FK → question.id)

Enrollment & Progress

- **enrollment**
 - id (PK), progress, student_id (FK → student_profile.id), course_id (FK → course.id)
- **student_progress**
 - id (PK), student_id (FK → student_profile.id), course_id (FK → course.id), completion, quiz_average, assignment_average

Security & Policies

- **Row-Level Security (RLS)** can be applied to sensitive tables:
 - enrollment, quiz_attempt, assignment_submission, student_progress.
- Students → only their own records.
- Instructors → only their courses.
- Administrators → full access.